

The two controllers that best fit the criteria are **Ryu** and **POX** (from the NOX/POX family).

1. Ryu

- **Python-Based:** Ryu is written entirely in 100% pure Python, making it accessible, lightweight, and easy to extend or integrate with external scripts and APIs.
- **Strong Community:** Backed by NTT with a broad global community, offering clear documentation, sample applications, and widespread industry/academic adoption.
- **Learning & Prototyping:** Its component-based architecture allows rapid development of control logic and REST APIs. Furthermore, it supports OpenFlow 1.0 through 1.5, NETCONF, and OF-Config, which is essential for interoperability in brownfield deployments containing legacy and modern devices.

2. POX (NOX/POX)

- **Python-Based:** While NOX was originally C++-based, POX was created as its pure Python successor.
- **Learning & Prototyping:** POX was explicitly designed as an educational and research platform for fast-iteration prototyping with a low barrier to entry and minimal code overhead.
- **Strong Community Legacy:** Developed and maintained through Stanford and academic channels, it features extensive tutorials and a deep repository of community examples suitable for testing basic OpenFlow concepts on brownfield networks.

Reasons for Excluding Other Candidates:

- **Floodlight:** Written entirely in Java, failing the Python requirement.
- **Trema:** Written in C and Ruby, failing the Python requirement.